

Species-Aware Module Review:

Peptidoglycan (MurNAc) Recycling in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI txid 160488; proteome UP000000556)

Review topic: `peptidoglycan_recycling` **Nominal bucket:** KEGG `ppu00520` — *Amino sugar and nucleotide sugar metabolism* (module area: `nucleotide_metabolism`) **Prepared for:** manual module satisfiability + gene-annotation curation

1. Executive summary

P. putida KT2440 possesses a **complete, biochemically validated cell-wall-sugar (MurNAc) recycling module** built around the *Pseudomonas*-defining **AmgK–MurU shortcut** that regenerates UDP-MurNAc directly from recycled N-acetylmuramic acid, **bypassing the de novo MurA/MurB route** (and thereby conferring intrinsic fosfomycin resistance) (Gisin et al. 2013,

P 23831760

Two of the committed enzymes — **AmgK (PP_0405)** and **MurU (PP_0406)** — were characterized *directly in P. putida*, and the downstream phosphatase **MupP (PP_1764)** was *first discovered in P. putida* (Borisova et al. 2017,

P 28351914

This is therefore an unusually strong, species-direct module.

Key curation conclusions:

- **The review topic is a focused sub-module of the broad KEGG map.** Only ~5 of the 25 candidate genes are true peptidoglycan-recycling steps (`amgK`, `murU`, `anmK`, `mupP`, `nagZ`). The remaining genes belong to **de novo UDP-GlcNAc/UDP-MurNAc synthesis** or to **unrelated nucleotide-sugar branches** (alginate, dTDP-rhamnose, UDP-glucose/galactose) that share map `ppu00520` and neighbors but are **not** recycling.

- **Several recycling steps are missing from the candidate metadata but are genomically present** and should be added: the importer **AmpG (PP_1355)**, the cytoplasmic recycling amidase **AmpD (PP_0789, EC 3.5.1.28)**, and the murein-peptide ligase **Mpl (PP_0547, EC 6.3.2.45)**. (Verified via KEGG ortholog links and UniProt proteome UP000000556.) Note the recycling amidase is **AmpD (PP_0789)**, distinct from the septal division amidase AmiC (PP_4897).
- **The *E. coli* catabolic branch is absent.** No **MurQ etherase (K07106)** gene exists in `ppu`; consequently the *P. putida* route is **anabolic (wall → wall)** rather than catabolic (wall → central metabolism). `murQ` should be marked **not_expected_in_target_taxon**.

Evidence strength: **strong/direct** for AmgK, MurU, MupP; **strong** (UniProt pathway tag + KEGG ortholog + *Pseudomonas* functional literature) for AnmK, NagZ, AmpG, AmpD, Mpl; **inference/not-assessed** only for accessory peptide-recycling steps not in scope here (e.g. LdcA LD-carboxypeptidase, peptide permeases).

Independent curation-grade confirmation (UniProt UP000000556): AmgK (PP_0405), MurU (PP_0406) and MupP (PP_1764) are annotated `PATHWAY: Cell wall biogenesis; peptidoglycan recycling` with **experimental ECO codes** — `ECO:0000269|PubMed:23831760` (AmgK, MurU) and `ECO:0000269|PubMed:28351914` (MupP); AnmK (PP_0434) and Mpl (PP_0547) carry the same recycling pathway tag; the importer is explicitly `Muropeptide permease AmpG` (PP_1355) and the amidase `1,6-anhydro-N-acetylmuramyl-L-alanine amidase AmpD` (PP_0789, cytoplasmic, Zn). Curators can therefore rely on UniProt pathway tags + ECO codes, not just homology.

2. Target-organism pathway definition

Included process (this module): the intracellular salvage of the amino sugar **N-acetylmuramic acid (MurNAc)** and its anhydro form (1,6-anhydro-MurNAc) that are released during normal peptidoglycan (PGN) turnover, converting them back into the activated cell-wall precursor **UDP-MurNAc** for reincorporation into peptidoglycan. In *P. putida* this proceeds:

PGN turnover → periplasmic lytic transglycosylases generate **GlcNAc-1,6-anhydro-MurNAc-peptides** → import (**AmpG**) → cytoplasmic **NagZ** removes GlcNAc → **AmpD** removes the stem peptide → **AnmK** phosphorylates 1,6-anhydro-MurNAc to MurNAc-6-P → **MupP** dephosphorylates to MurNAc → **AmgK** (anomeric C1 kinase) → MurNAc-1-P → **MurU** → **UDP-MurNAc**.

Neighboring processes to keep separate:

- **De novo UDP-GlcNAc biosynthesis** (GlmS→GlmM→GlmU) and **de novo UDP-MurNAc synthesis** (MurA→MurB). These *feed the same product pool* but are a different module; the recycling shortcut explicitly *bypasses* MurA/MurB.
- **Peptidoglycan biosynthesis proper** (MurC–F, MurG, MraY, PBPs) — downstream consumers of UDP-MurNAc.
- **Other nucleotide-sugar metabolism in `ppu00520`/neighbors**: alginate/GDP-mannose (`algA`, `algC`, `algD`, `cpsG`, `PP_1776`), dTDP-rhamnose (`rfbA`), UDP-glucose/UDP-galactose/UDP-glucuronate (`galU`, `galE`, `udg`, `pgm`, `pgi1/2`, `glk`, `PP_0501`). None are peptidoglycan recycling.
- **Enterobacterial-common-antigen / UDP-ManNAc branch** (`rffE` / `wecB`, UDP-GlcNAc 2-epimerase) — a UDP-GlcNAc consumer, not recycling.

Alternate names / database definitions: "PGN recycling", "cell-wall sugar recycling/salvage", "MurNAc recycling", "AmgK/MurU recycling shortcut", "muropeptide recycling". KEGG folds the enzymes into map 00520; there is no dedicated KEGG *pathway* for the *Pseudomonas* recycling shortcut, so a bespoke module document is appropriate.

3. Expected step model

#	Step (function)	Enzyme/type	EC	Expected?	Notes
S0	Generate 1,6-anhydro-MurNAc muropeptides (turnover)	Lytic transglycosylases (Slt/MltF)	4.2.2.29	Yes	Upstream of module; many paralogs
S1	Import muropeptides across inner membrane	AmpG MFS permease	–	Yes	Missing from metadata
S2	Remove terminal GlcNAc	NagZ β -N-acetylglucosaminidase	3.2.1.52	Yes	In metadata (bucket ppu01501)
S3	Remove stem peptide	AmpD amidase (PP_0789)	3.5.1.28	Yes	Missing from metadata; distinct from septal AmiC PP_4897
S4	Phosphorylate 1,6-anhydro-MurNAc $\rightarrow$ MurNAc-6-P	AnmK	2.7.1.170	Yes	In metadata
S5	Dephosphorylate MurNAc-6-P $\rightarrow$ MurNAc	MupP (HAD)	3.1.3.105	Yes	In metadata; <i>first found in P. putida</i>
S6	Anomeric (C1) phosphorylation MurNAc $\rightarrow$ MurNAc-1-P	AmgK	2.7.1.221	Yes	In metadata; direct <i>P. putida</i> evidence
S7	Uridylylation MurNAc-1-P $\rightarrow$ UDP-MurNAc	MurU	2.7.7.99	Yes	In metadata; direct <i>P. putida</i> evidence
—	<i>Catabolic diversion</i> MurNAc-6-P $\rightarrow$ GlcNAc-6-P	MurQ etherase	4.2.1.126	No	Absent in <i>Pseudomonas</i> ; makes route anabolic

#	Step (function)	Enzyme/type	EC	Expected?	Notes
S8	Re-ligate intact stem peptide onto UDP-MurNAc	Mpl (PP_0547)	6.3.2.45	Yes	Confirmed in UniProt UP000000556 (see §5)

4. Candidate genes and evidence

4a. Core recycling steps — COVERED (promote to full review)

- **amgK** — PP_0405 | Q88QT3 | EC 2.7.1.221 (K07102). Anomeric MurNAc/GlcNAc 1-kinase; makes MurNAc- α -1-P. **Direct P. putida biochemical characterization** (Gisin 2013). Committed step. *Caveat*: ROK/kinase family; the UniProt "GlcNAc kinase" side-activity broadens EC/GO mapping — annotate primarily as the C1 MurNAc kinase.
- **murU** — PP_0406 | Q88QT2 | EC 2.7.7.99 (K00992). MurNAc- α -1-P uridylyltransferase; makes UDP-MurNAc. **Direct P. putida evidence** (Gisin 2013). Adjacent to *amgK* (likely operon). Committed step.
- **anmK** — PP_0434 | Q88QQ4 | EC 2.7.1.170 (K09001). Anhydro-MurNAc kinase; funnels the *actual* turnover product (1,6-anhydro-MurNAc) into the route. KEGG ortholog + Borisova 2017 model. High confidence.
- **mupP** — PP_1764 | Q88M11 | EC 3.1.3.105 (K22292). MurNAc-6-P phosphatase; the essential link that connects AnmK output to AmgK input. **First characterized in P. putida** (Borisova 2017). *Caveat*: KEGG gene NAME is the generic "Phosphoglycolate phosphatase 2" and UniProt lists the gene synonym **gph** (HAD superfamily) — an **over-general annotation** that hides its verified recycling role; curate to EC 3.1.3.105 (UniProt already tags it peptidoglycan recycling, ECO:0000269 | PubMed:28351914).
- **nagZ** — PP_2145 | Q88KZ4 | EC 3.2.1.52 (K01207). Cytoplasmic β -N-acetylglucosaminidase removing GlcNAc from imported muropeptides. Functionally validated across *Pseudomonas* as a recycling enzyme and drug target (Torrens 2019

P 31325363

Ho 2018

P 30178799

Caveat: its metadata **primary bucket is ppu01501 (β -lactam resistance)**, not ppu00520 — genuine recycling step nonetheless; the broad EC 3.2.1.52 also covers other β -hexosaminidases.

4b. De novo UDP-sugar synthesis — present but a different module (keep separate)

- **glmS PP_5409 (EC 2.6.1.16), glmM PP_4716 (EC 5.4.2.10), glmU PP_5411 (EC 2.3.1.157/2.7.7.23)** — de novo UDP-GlcNAc. Real, but *biosynthetic*, not recycling.
- **murA PP_0964 (EC 2.5.1.7), murB PP_1904 (EC 1.3.1.98)** — de novo UDP-MurNAc. **These are exactly the steps the AmgK–MurU shortcut bypasses**; their presence does not satisfy the recycling module and vice-versa.

4c. Candidates that are out of scope for peptidoglycan recycling

rffE/PP_1811 (UDP-GlcNAc 2-epimerase, ECA/UDP-ManNAc), algA/PP_1277, PP_1776, algD/PP_1288, algC/PP_5288, cpsG/PP_1777 (alginate/GDP-mannose), rfbA/PP_1783 (dTDP-rhamnose/LPS), galU/PP_3821, udg/PP_2926, galE/PP_3129, pgm/PP_3578, pgi1/PP_1808, pgi2/PP_4701, glk/PP_1011, PP_0501. These co-occur in the broad amino/nucleotide-sugar map but are **not** peptidoglycan recycling and were pulled in by resolving the whole ppu00520 bucket.

5. Gaps, ambiguities, and likely over-annotations

Missing-from-metadata but genomically present (gap-fill):

- **AmpG importer — PP_1355 (K08218).** Inner-membrane MFS permease for GlcNAc-anhMurNAc muropeptides; essential first cytoplasmic step. *Add to module.* (Functional importance shown in *P. aeruginosa*; ortholog transfer strong.)
- **AmpD amidase — PP_0789 (Q88PQ9, EC 3.5.1.28).** "1,6-anhydro-N-acetylmuramyl-L-alanine amidase AmpD" — the *cytoplasmic* recycling amidase releasing the stem peptide from anhydro-muropeptides. *Add to module.* **Do not confuse with the septal/division amidase AmiC (PP_4897)** or the periplasmic AmiD (PP_0130); KT2440 has ≥ 4 amidase paralogs (PP_0130, PP_0789, PP_2269, PP_4897) with distinct roles — only PP_0789 is the core recycling AmpD.

- **Mpl murein peptide ligase — PP_0547 (Q88QE7, EC 6.3.2.45).** Re-ligates the intact L-Ala- γ -D-Glu-*meso*-DAP tripeptide directly onto UDP-MurNAc, enabling *peptide-level* recycling. *Add to module.* (This resolves the earlier candidate_uncertain call; the KO-only probe had missed it.)
- **Lytic transglycosylases** (e.g. PP_2130 Slt K08309, PP_1036 MltF K18691) generate the 1,6-anhydro-MurNAc substrate. Upstream context; large paralog family, do not force single-gene mapping.

Not expected in this taxon:

- **MurQ etherase (K07106): absent in `ppu`.** No gene links to K07106. This is the mechanistic reason the *Pseudomonas* route recycles MurNAc back to the wall rather than diverting MurNAc-6-P to GlcNAc-6-P/central metabolism as in *E. coli*. Mark `murQ` **not_expected_in_target_taxon.**

Resolved (was ambiguous):

- **Mpl (murein peptide ligase, EC 6.3.2.45): present as PP_0547 (UniProt Q88QE7).** Peptide-level recycling is therefore intact. The earlier "no ortholog" impression came from a KO-only probe (K05363 not linked); the authoritative UniProt annotation confirms presence. Lesson for curation: KO-link absence alone is not evidence of absence — confirm by protein/EC search. (My first automated KEGG probe also mis-labeled K01925 as Mpl when it is MurD, and K18691 as AmpG when it is MltF — KO-only calls need sequence confirmation.)

Likely over-annotations / broad mappings:

- **MupP/PP_1764** carries a generic HAD "phosphoglycolate phosphatase" name (see §4a).
- **NagZ/PP_2145** EC 3.2.1.52 is a broad β -hexosaminidase class; primary bucket mis-placed in `ppu01501`.
- **AmgK** "GlcNAc kinase" secondary activity broadens its GO/EC footprint.

6. Module and GO-curation recommendations

Module step	Status	Gene(s)
Muropeptide import	covered (add)	AmpG PP_1355
GlcNAc removal	covered	NagZ PP_2145
Peptide removal (amidase)	covered (add)	AmpD PP_0789 (not AmiC PP_4897)
anhMurNAc → MurNAc-6-P	covered	AnmK PP_0434
MurNAc-6-P → MurNAc	covered	MupP PP_1764
MurNAc → MurNAc-1-P	covered	AmgK PP_0405
MurNAc-1-P → UDP- MurNAc	covered	MurU PP_0406
MurNAc-6-P → GlcNAc-6-P (catabolic)	not_expected_in_target_taxon	MurQ (none)
Stem-peptide re-ligation	covered (add)	Mpl PP_0547
de novo UDP-MurNAc (MurA/MurB)	module_needs_revision — separate module	PP_0964, PP_1904

Boundary verdict: the generic `ppu00520` bucket is **too broad** for this review. Recommend a **dedicated "peptidoglycan/MurNAc recycling (AmgK–MurU shortcut)" module** for *Pseudomonas*, containing AmpG (PP_1355), NagZ (PP_2145), AmpD (PP_0789), AnmK (PP_0434), MupP (PP_1764), AmgK (PP_0405), MurU (PP_0406), and Mpl (PP_0547); explicitly excluding MurA/MurB (de novo) and the unrelated nucleotide-sugar genes; and explicitly recording MurQ as **absent** in the lineage.

GO-curation flags: the newer EC activities — **AmgK 2.7.1.221**, **MurU 2.7.7.99**, **MupP 3.1.3.105**, **AnmK 2.7.1.170** — should be checked for precise GO molecular-function terms; MupP's specific "MurNAc-6-phosphate phosphatase activity" and MurU's "MurNAc-1-phosphate uridylyltransferase activity" are the most likely to need term requests or tightened mappings. Recommend GO annotations with experimental evidence codes for AmgK/MurU/MupP citing PMIDs 23831760 and 28351914.

7. Genes to promote to full fetch-gene review

1. **PP_0405 amgK, PP_0406 murU, PP_1764 mupP** — direct *P. putida* experimental evidence; highest-value, curate with EXP GO codes.
2. **PP_0434 anmK, PP_2145 nagZ** — confirm EC/GO and fix bucket placement (NagZ).
3. **PP_1355 (AmpG), PP_0789 (AmpD), and PP_0547 (Mpl)** — add to module. For AmpD, distinguish from the amidase paralogs AmiD (PP_0130), AmiC (PP_4897), and PP_2269, which serve periplasmic/septal (not recycling) roles.

8. Key references

- **Gisin, Schneider, Nägele, Borisova, Mayer (2013).** *A cell wall recycling shortcut that bypasses peptidoglycan de novo biosynthesis.* Nat Chem Biol.

P 23831760

— Direct characterization of AmgK & MurU in *P. putida*; intrinsic fosfomycin resistance.

- **Borisova, Gisin, Mayer (2017).** *[Blocking peptidoglycan recycling / MurNAc recycling routes; MupP characterization].* mBio.

P 28351914

— MupP (MurNAc-6-P phosphatase) first described in *P. putida*; two recycling routes.

- **Torrens, Sánchez-Diener, Jordana-Lluch, Barceló, Zamorano, Juan, Oliver (2019).** *In vivo validation of peptidoglycan recycling as a target...*

P 31325363

— AmpG/NagZ recycling functional in *Pseudomonas*.

- **Ho, Winogradzki, Debowski, Madden, Voadlo, Mark, Stubbs (2018).** *Mechanism-based NagZ inactivator reverses β -lactam resistance in *P. aeruginosa*.*

P 30178799

- **Barceló et al. (2022)**

P 35171032

Juan et al. (2017)

P 29029112

— recycling–resistance–virulence interplay (broader *Pseudomonas*).

- **KEGG / UniProt** for `ppu` (rest.kegg.jp; UniProt UP000000556): amgK PP_0405/K07102; murU PP_0406/K00992; anmK PP_0434/K09001; mupP PP_1764/K22292; nagZ PP_2145/K01207; ampG PP_1355/K08218; **ampD PP_0789** (Q88PQ9, EC 3.5.1.28); **mpl PP_0547** (Q88QE7, EC 6.3.2.45); murQ (K07106) — no `ppu` gene. Amidase paralogs: AmiD PP_0130, AmiC PP_4897, PP_2269.

Uncertainty & species-transfer notes

- **Direct for KT2440:** AmgK, MurU (enzymology in *P. putida*); MupP (discovery in *P. putida*).
- **Strong transfer / ortholog-confirmed:** AnmK, NagZ, AmpG, AmpD (KEGG orthologs in `ppu`; functional data largely from *P. aeruginosa*, same PGN chemistry).
- **Regulatory caveat:** the recycling→AmpC- β -lactamase induction circuit is well established in *P. aeruginosa*; *P. putida* KT2440 is environmental and its β -lactamase regulation is less characterized, so do **not** transfer the resistance-induction phenotype uncritically to KT2440. The *metabolic* recycling module, however, transfers well.
- **Open experiments / expert questions:** (i) test whether $\Delta mupP$ or $\Delta amgK$ abolishes MurNAc recycling and restores fosfomycin susceptibility in KT2440 *specifically* (current fosfomycin-resistance rationale is from the heterologous *E. coli* engineering in Gisin 2013); (ii) confirm AmpG (PP_1355) is the physiological muropeptide importer in KT2440 vs. an Opp-type peptide transporter; (iii) clarify the division of labour among the four amidase paralogs (PP_0130/PP_0789/PP_2269/PP_4897); (iv) determine whether the environmental strain KT2440 couples recycling to any β -lactamase induction, given its divergence from clinical *P. aeruginosa*.